ORGANISMS & THE ENVIRONMENT

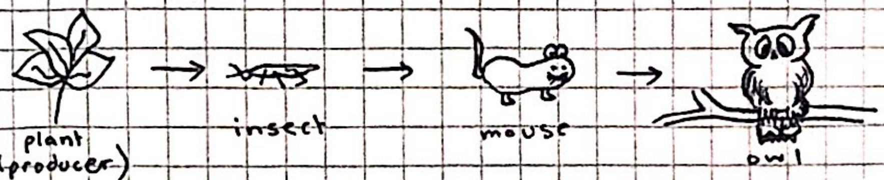
23/01/23

organisms & the environment

* the sun is the principal source of energy input to biological systems

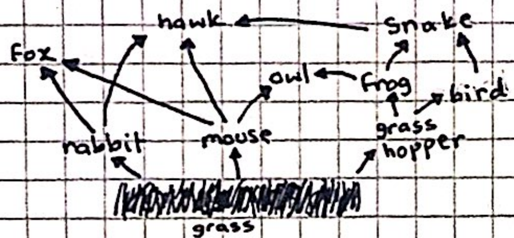
energy flows through living organisms
L including light energy from sun and chemical energy in organisms
L energy is eventually trasferred to the environment

Food chain:
shows the transfer of energy from one organism to another/the next, beginning with a producer.



} food chain showing how energy is transferred between organisms

Food web:
a network of interconnected food chains



} food web

Producer:
an organism that makes its own organic nutrients, usually using energy from sunlight, through photosynthesis

Consumer:
an organism that gets its energy by feeding on other organisms
L maybe classed as primary, secondary, tertiary & quatenory (in food chain

Herbivore:
an animal that gets its energy by eating plants

Carnivore:
an animal that gets its energy by eating other animals

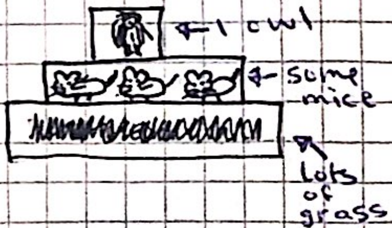
Decomposer:
an organism that gets its energy from dead or waste organic material

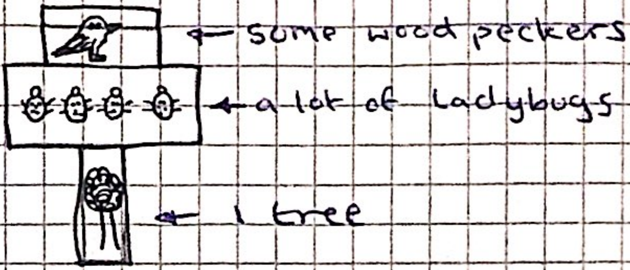
* most of the changes in populations of animals and plants happen as a result of human impact — either by overharvesting of food species or by the introduction of foreign species to a habitat.

Pyramids of numbers:

shows how many organisms we are talking about at each level of a food chain

L the width of the box indicates the number of organisms at that trophic level

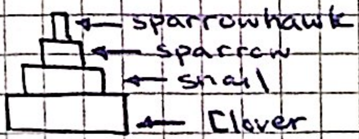




Pyramids of biomass:

shows how much mass the creatures at each level would have without including all the water that is in the organisms (dry mass)

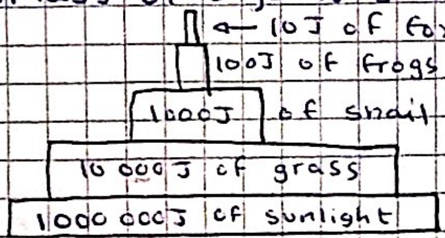
L always pyramid-shaped



- the pyramid of biomass provides a much better idea of the quantity of the plant/animal material at each level of a food chain and therefore a better representation of the interdependancy.

Pyramids of energy:

shows the total quantity of available energy stored in the biomass of organisms at each level



- the pyramid of energy provides a much better idea of the actual amount of energy gained by the the next trophic level

Trophic Level:

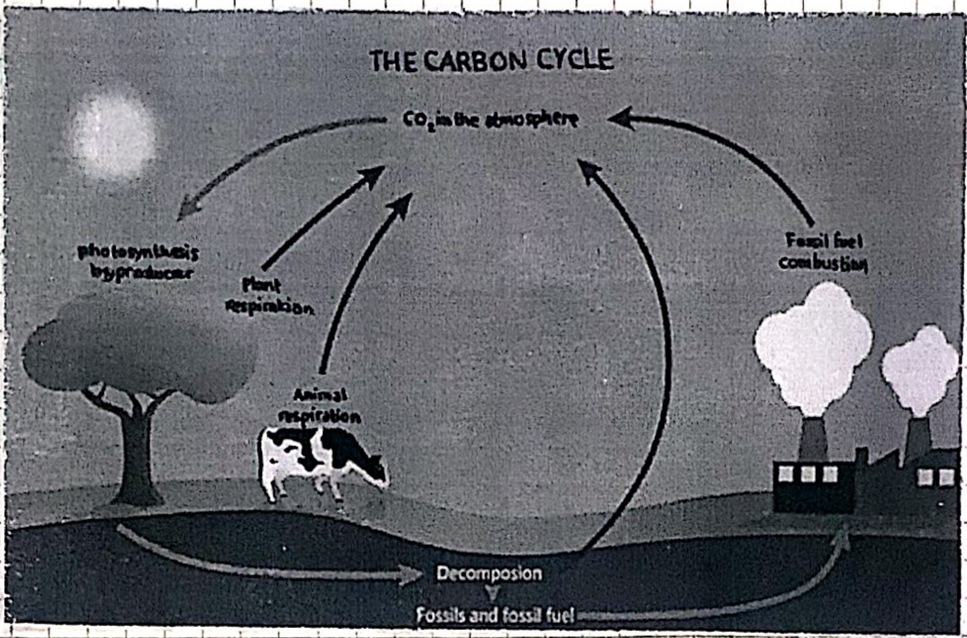
the position of an organism in a food chain, food web or ecological pyramid.

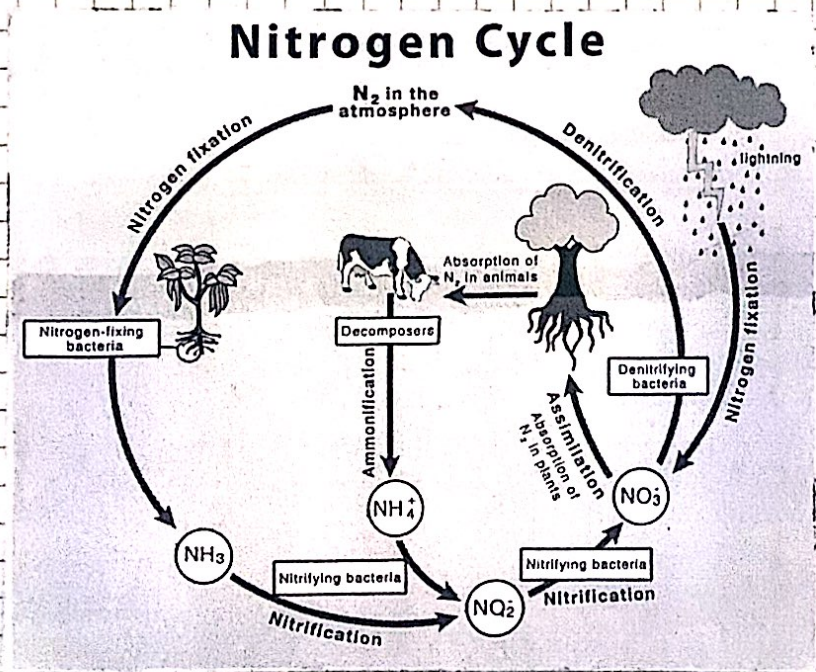
→ producers, primary consumers, secondary consumers, tertiary consumers and quaternary consumers are defined as the trophic levels of food webs, food chains & ecological pyramids

* majority of the energy an organism recieves gets lost or used:
  - L making waste products
  - L as movement
  - L as heat
  - L as undigested waste

- the inefficient loss of energy explains why food chains are usually fewer than 5 trophic levels.

wheat → cow → humans
wheat → humans
↳ it is clear that there is much more energy available to humans by eating the wheat rather than the cow that ate wheat.





- the plants/trees produce amino acids and protein which animals then feed and digest. They later deaminate it.
- nitrogen is required to make proteins

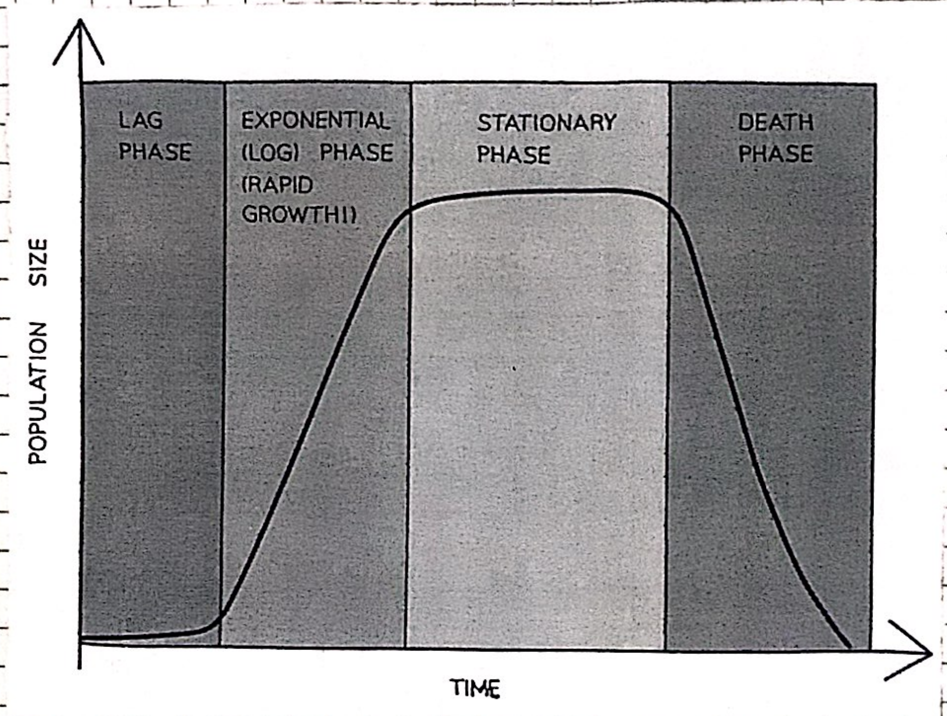
- when animals/plants die, they decay and all the proteins inside them are broken down in ammonium compounds and put back into the soil by decomposers.
- plants can't absorb ammonium compounds though, so a second type of soil bacteria, nitrifying bacteria, converts the ammonium compounds to nitrites then to nitrates for the plant.
- nitrogen-fixing bacteria fix atmospheric nitrogen to ammonia and then convert it into nitrites and nitrates which can then be taken by the plant.
- the unhelpful type of anaerobic bacteria is called dentrifying bacteria found in poorly aerated soil. It takes the nitrate out of the soil and converts them back to N_2 gas.

Population:-
a group of organisms of one species, living in the same area, at the same time

Community:-
all of the populations of different species in an ecosystem

Ecosystem:-
a unit containing the community of organisms and their environment, interacting together

* population growth in most organisms is controlled by:-
  - L food supply
  - L predation
  - L disease



Lag phase = organisms are adapting to their environment before they are able to reproduce. At this stage there are very few organisms thus small number of offspring

Log phase = growth is exponential (rapid). Food supply is abundant, birth rate is rapid and death rate is low.

stationary phase = population levels out due to a factor in the environment, such as limited nutrients. Birth rate and death rate are equal and will remain so until the nutrient is replenished or becomes severely limited

death phase = population decreases as death rate is now greater than birth rate, usually because food supply is short or metabolic waste produced by the population have built up to toxic levels.